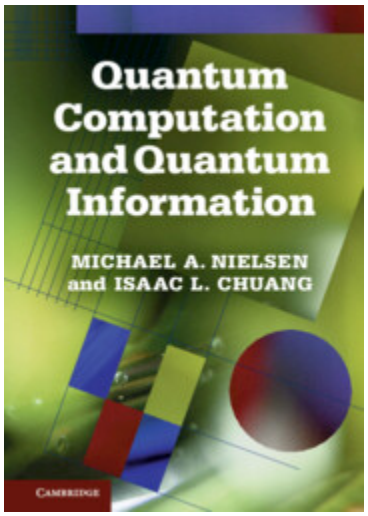
Quantum Computation and Quantum Information



10th Anniversary Edition

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Physical publication date: 31 January 2011

Find out more

One of the most cited books in physics of all time, Quantum Computation and Quantum Information remains the best textbook in this exciting field of science. This 10th anniversary edition includes an introduction from the authors setting the work in context. This comprehensive textbook describes such remarkable effects as fast quantum algorithms, quantum teleportation, quantum cryptography and quantum error-correction. Quantum mechanics and computer science are introduced before moving on to describe what a quantum computer is, how it can be used to solve problems faster than 'classical' computers and its real-world implementation. It concludes with an in-depth treatment of quantum information. Containing a wealth of figures and exercises, this well-known textbook is ideal for courses on the subject, and will interest beginning graduate students and researchers in physics, computer science, mathematics, and electrical engineering.

Key features

- The best introduction to quantum computing and quantum information, written by experts on the subject
- Gives a comprehensive introduction to the main ideas and techniques, with hundreds of exercises and figures
- Contains extensive background material so it can be understood without prior knowledge of quantum mechanics or quantum science

About the book

Subjects: Computer Science, Cryptography, Cryptology and Coding, Physics and Astronomy, Quantum Physics and Quantum Information

Format: Hardback , Price: US$89.00
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